

Importance of Water

- Water is one of the most essential element needed by man, plants and animals for their survival. **No water, no life!**
- Water maintains an ecological balance-**a balance in the relationship between living things and the environment they live in.**
- Life in an organized community require the twin services of water supply and sewage disposal.
- Good sanitation cannot be maintained without adequate water supply system
- The use of water is increasing rapidly, with increase in population. Already, there are acute shortages of both surface and underground waters in many parts of the world.
- Careless pollution and contamination of streams, lakes, reservoirs, wells and other underground sources will greatly impair the quality of available waters
- This pollution results because of improper disposal of waste water-both domestic and industrial.

Objectives of water supply system

- To provide wholesome water to the consumer for drinking purpose.
- To supply adequate quantity of water to meet the least minimum needs of the individuals.
- To make adequate provisions for emergencies like fire-fighting, festivals, meeting, etc.
- To make provisions for future demand due to increase in population, increase in standard of living, storage and conveyance.
- To prevent pollution of water at source, storage and conveyance.
- To maintain the treatment units and distribution system in good condition with adequate staff and material
- To design and maintain the system that is economical and reliable



Sources of Raw Water

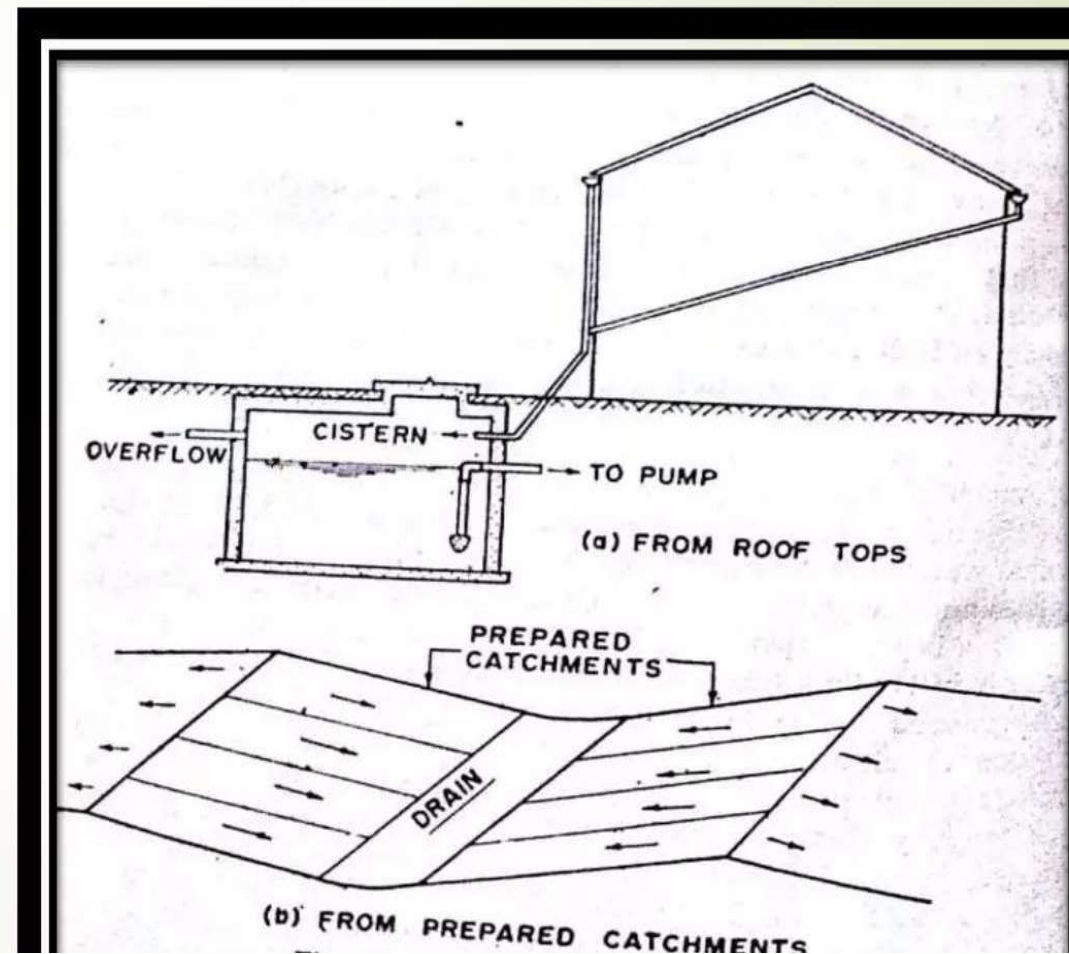
- ▶ The following are common sources of water that can be harnessed to meet water demand of the population.
 - i. Rain water
 - ii. Surface water
 - iii. Ground water
 - iv. Water obtained from reclamation.



Sources of Raw Water

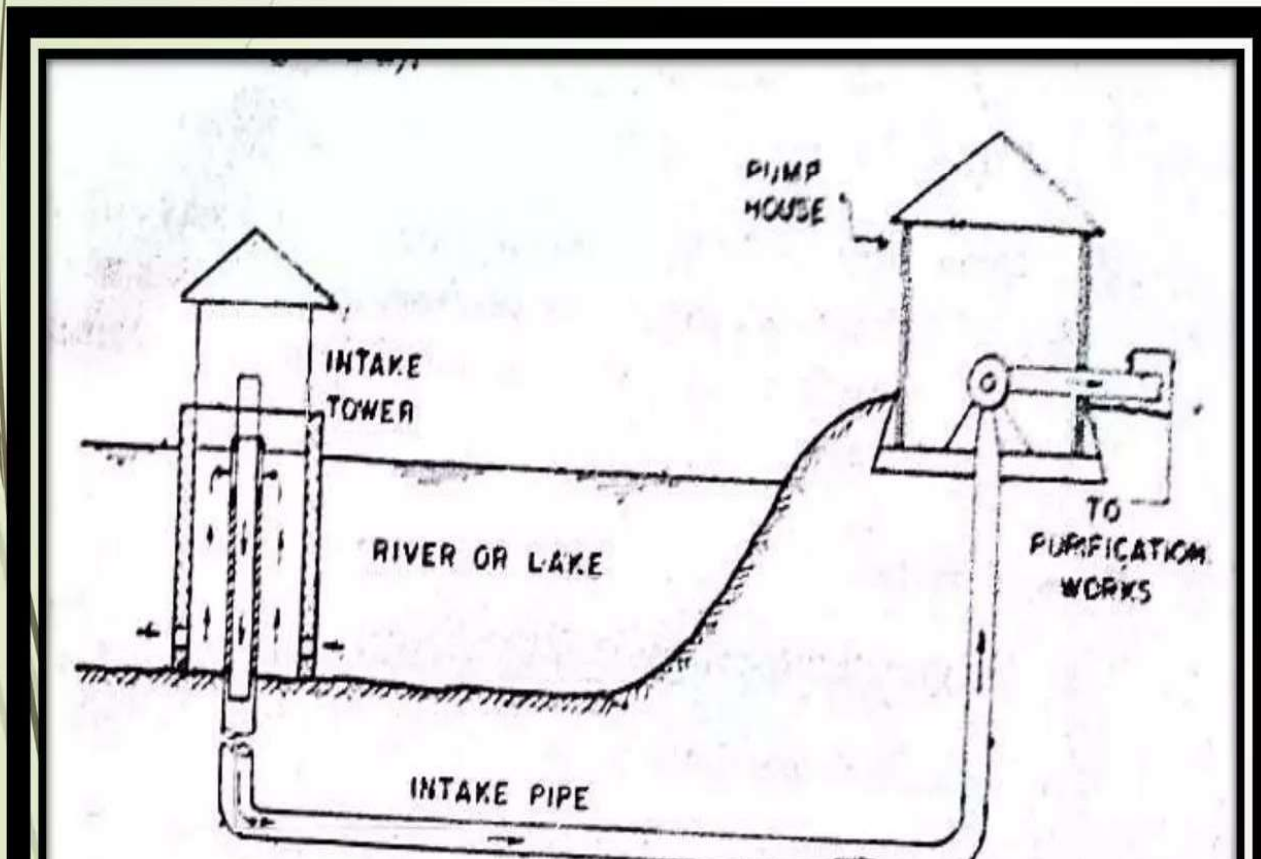
► Rain water:

- a. **From roof Tops:** Water is stored in small underground tanks or cistern, for small individual supply.
- b. **From prepared Catchments:** The surface of catchment is made impervious by suitable lining material, and suitable slope is given so that the water is stored in moderate size reservoirs. This water is used for communal supplies, mostly for drinking purposes.



Sources of Raw Water

► Surface water

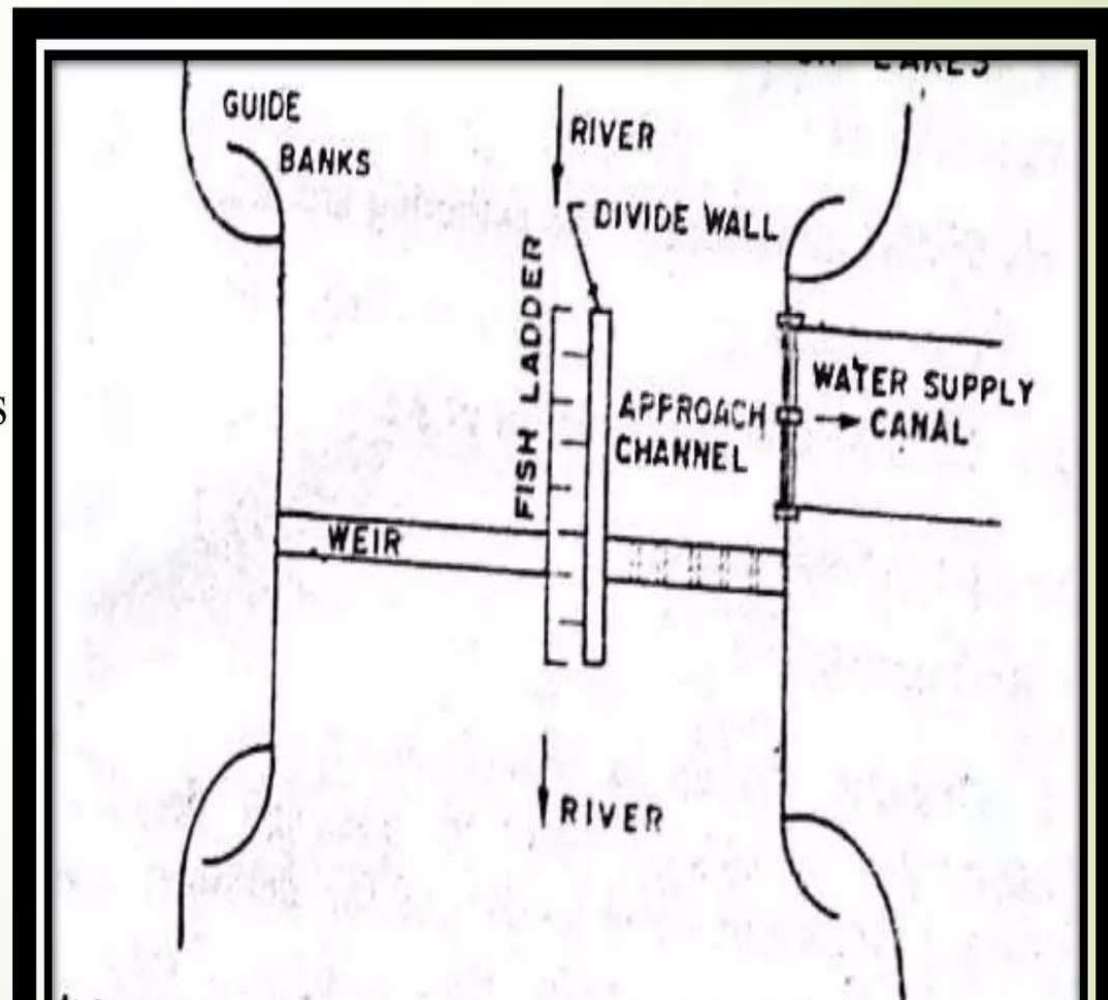


(a) From rivers by continuous draft: Water may be collected from river without any diversion work

Sources of Raw Water

► Surface water

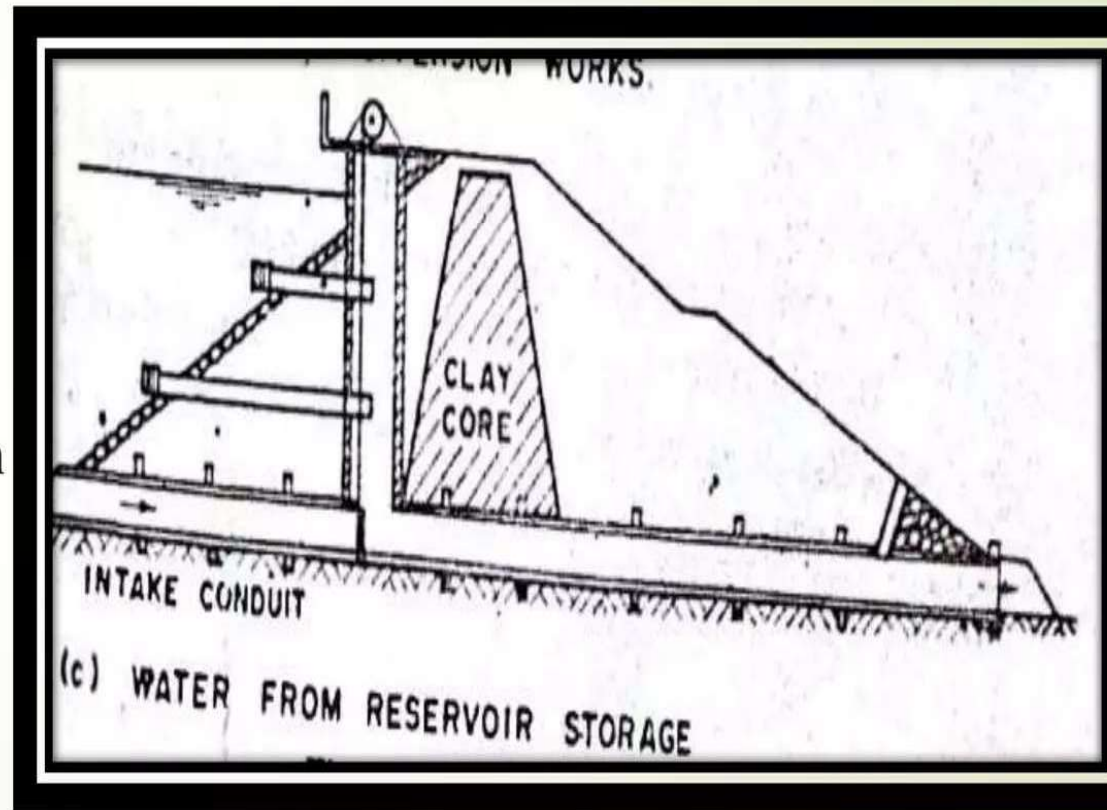
(b). From river diversion: A diversion work is constructed across a perennial river and water is diverted into a canal which leads water to the site of water purification works.



Sources of Raw Water

► Surface water

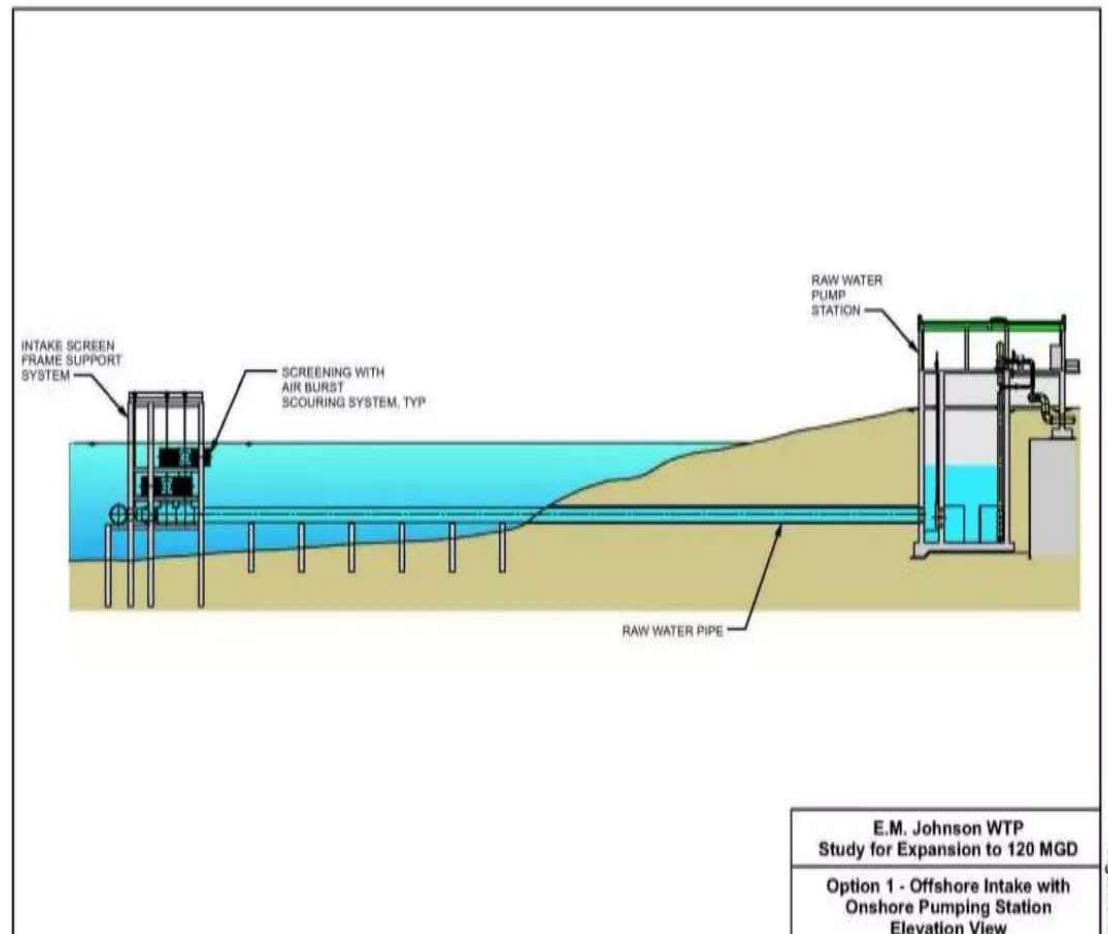
(c). From reservoir storage: Where supply is not ensured throughout the year, dam may be constructed across the river and water stored in the reservoir.



Sources of Raw Water

► Surface water

(d). From direct intake from natural lakes: Water may also be obtained from direct intake from natural lakes which receive surface run-off from the adjoining catchment.



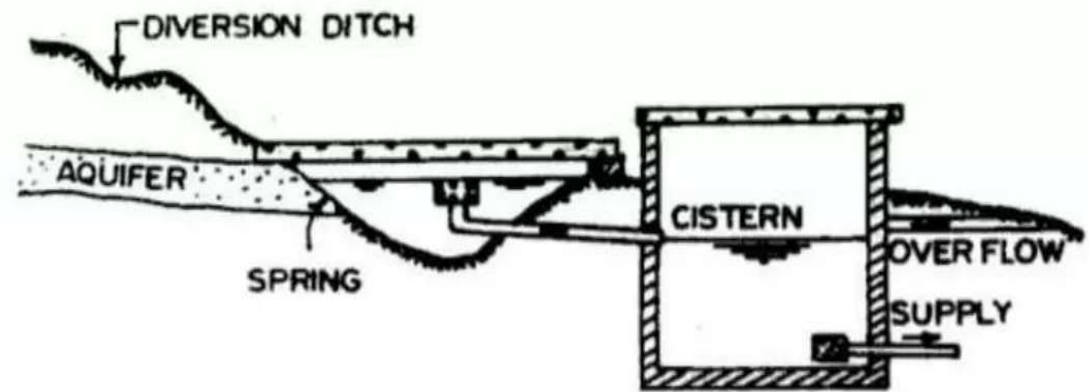
Sources of Raw Water

► Ground Water

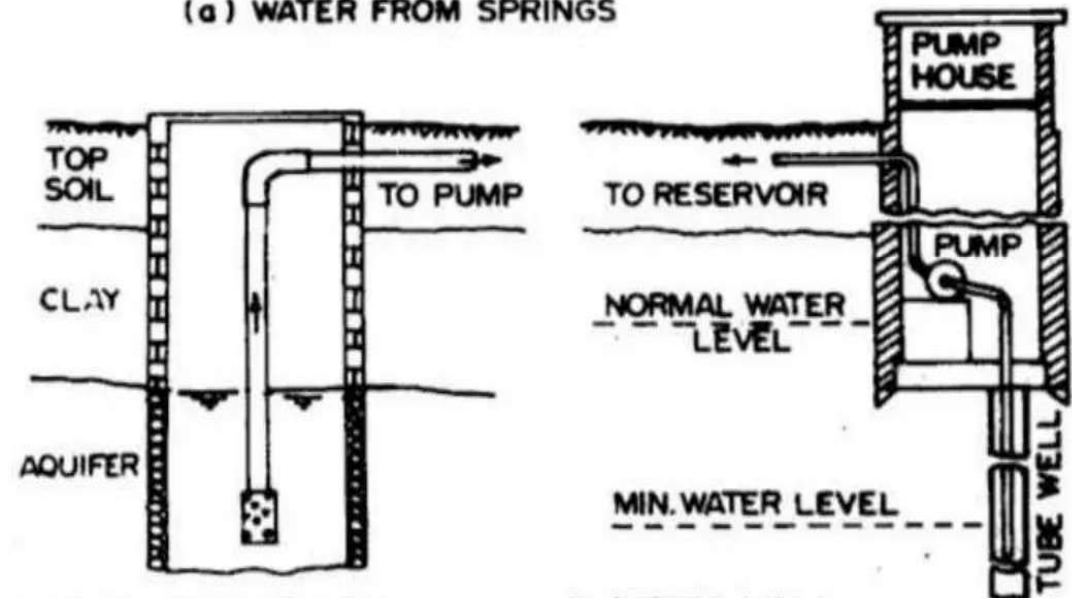
The ground water reservoir consists of water held in voids within a geologic stratum. The ground water can be tapped from the following sources;

- i. From natural springs
- ii. From wells and bore holes
- iii. From infiltration galleries, basins or cribs
- iv. From river side radial collector well

► Ground Water

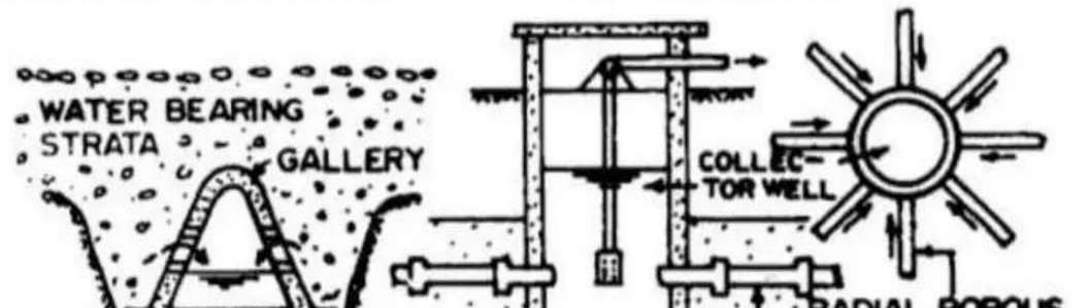


(a) WATER FROM SPRINGS



(b) SHALLOW DUG WELL

(bii) TUBE WELL



Sources of Raw Water

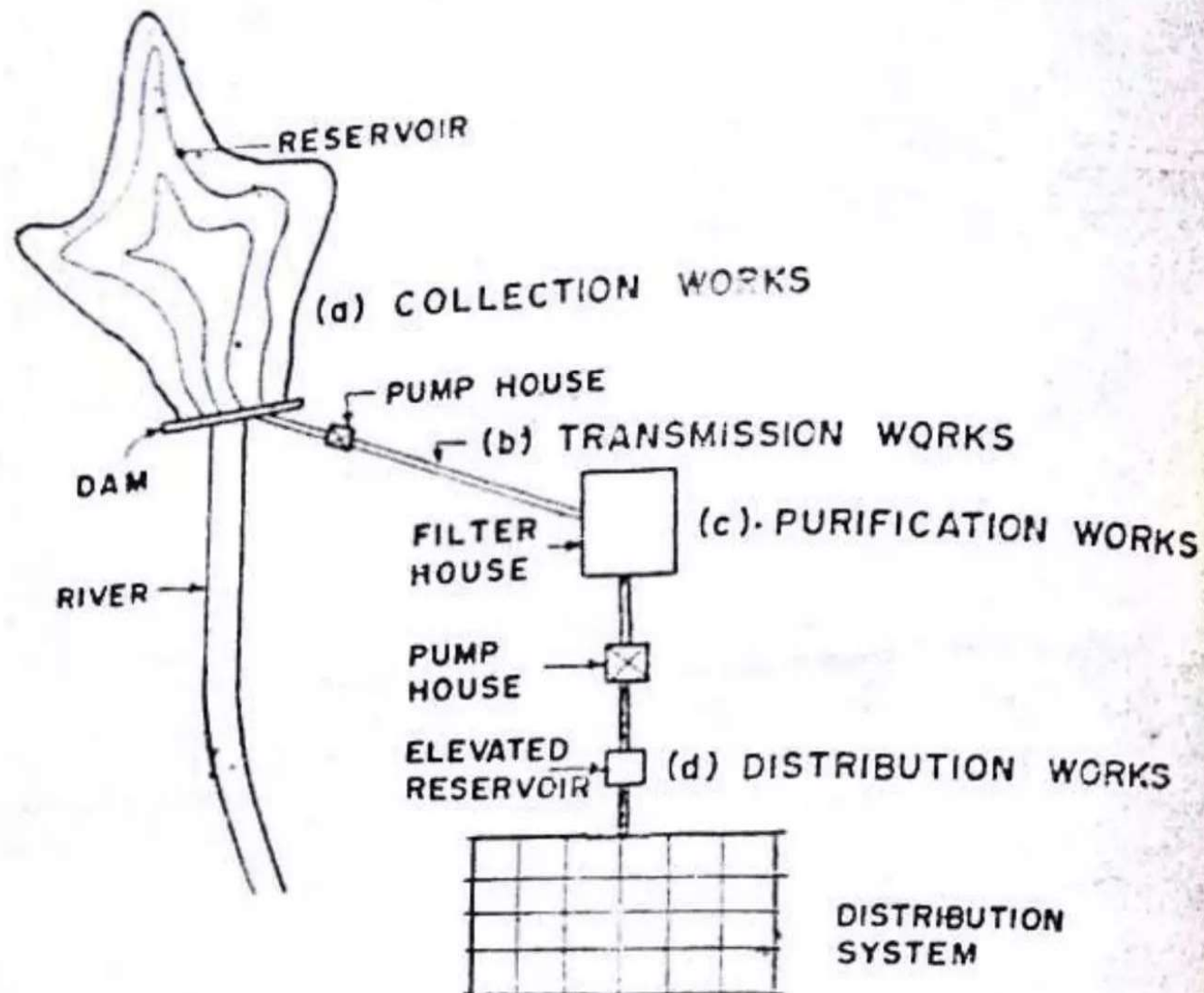
► Water Obtained by reclamation

- i. **Desalinization:** Saline or brackish water may be rendered useful for drinking purposes by installing desalinization plants. The common methods used for desalinization are; distillation, reverse osmosis, electrodialysis, freezing and solar evaporation.
- ii. **Re-use of treated waste water:** Effluent or waste water can be treated suitably so that it may be reused. An example of the controlled indirect reuse in the intentional artificial recharge of ground water aquifers by adequately treated wastewater.

Water Supply Systems

- The primary objective of water treatment for public supply is to take water from the best available source and subject it to processing which will ensure water of good physical quality, free from unpleasant taste or odor and containing nothing which may be detrimental to health.
- Municipal water supply systems consist of the following units
 - i. Collection works**
 - ii. Transmission works**
 - iii. Purification works and**
 - iv. Distribution works**

Water Supply Systems



Water Supply Systems

- **Collection works:** Intake structures that convey water to purification units. But if the water source is intermittent, a dam is built across the river so that water is stored in a reservoir. Water is then drawn from the reservoir as per needs: The collection works consist of storage or diversion work, and an intake structure.
- **Transmission works:** Collection may be far away from city to be supplied. In this case, water is conveyed to the city through transmission works. This forms the connecting link between the collection work and purification works. Transmission works may be in the form of conduits, canals or aqueducts. For gravity flow, canals are generally used.
- **Purification works:** Water collected directly from source may not be safe for drinking because of physical, chemical & biological impurities. The water works must deliver to the population water that is;
 - i. **Hygienically safe**

Water Supply Systems

- **Distribution works:** The treated and purified water is then sent to consumers through suitable distribution systems. In order that water may flow in the water pipes under pressure, the purified water is normally stored in an *elevated service reservoir*. More than one reservoir may be needed in a large system.
- There are two patterns of water distribution system:
 - i. **Branching pattern with dead ends**
 - ii. **Grid iron pattern**